AMC 10 Topical Test 6

Worked solutions · Choose a question to review.

Question 1 · Answer C

A map is enlarged so its area increases by 69%. A road that was 20 cm long on the map becomes how many centimetres long?

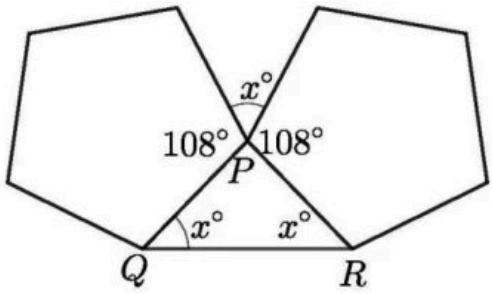
A. 29 **B.** 23 **C.** 26 **D.** 34 **E.** 32

Lengths scale by the square root of the area factor. The length factor is $\sqrt{1.69}=1.3$, and $20\times1.3=26$.

Question 2 · Answer C

SOLUTION **C**

Let P, Q and R be the points shown in the diagram.



The interior angles of a regular pentagon are all 108°.

Because the two pentagons are congruent, $PR = PQ$. Therefore $\angle PRQ = \angle PQR = x°$.

Because the sum of the angles in a triangle is 180°, from the triangle PQR, we have $x° + x° + \angle QPR = 180°$. Therefore $\angle QPR = 180° - 2x°$.

The sum of the angles at a point is 360°. Therefore from the angles at the point P, we have

$$(180° - 2x°) + 108° + x° + 108° = 360°.$$

That is

$$396° - x° = 360°.$$

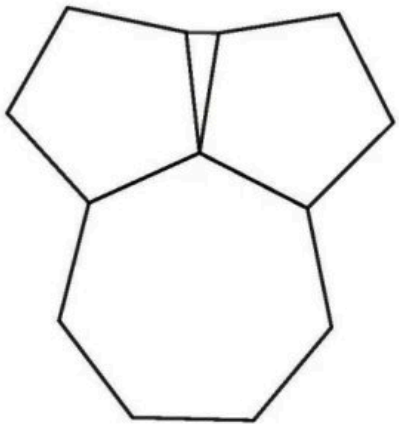
It follows that

$$x° = 396° - 360° = 36°.$$

FOR INVESTIGATION

- Prove that the sum of the angles in a triangle is 180°.
- Prove that each interior angle of a regular pentagon is 108°.
- Find a formula in terms of n for the size of the interior angles of a regular polygon with n sides.
- The diagram shows a regular heptagon, two regular pentagons and a triangle.

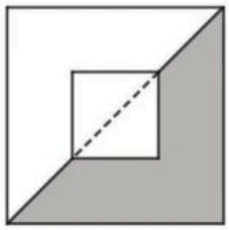
  What are the interior angles of the triangle?



Question 3 · Answer D

SOLUTION **D**

The small square has area 2^2, that is, 4. The shaded region is made up of half the large square with side length 5, with half the small square with side length 2 removed.



Therefore, the shaded area is $\frac{1}{2}(5^2) - \frac{1}{2}(2^2) = \frac{25}{2} - 2 = \frac{21}{2}$. Hence the ratio of the area of the small square to that of the shaded region is $4 : \frac{21}{2}$, that is, $8 : 21$.

FOR INVESTIGATION

- Suppose that the area of the shaded region were one-third of the area of the outer square. What would be the ratio of the area of the inner square to the area of the outer square?
- Suppose that the ratio of the shaded area to the area of the small square is $7 : 18$. What is the ratio of the side length of the small square to the side length of the large square?

Question 4 · Answer B

Three sides of a triangle have lengths 7,10 and x, where x is an integer. How many distinct values of x are possible?

A. 14 **B.** 13 **C.** 12 **D.** 16 **E.** 15

Strict triangle inequalities require $3<x<17$, so the integers 4 through 16 give 13 possibilities.

Question 5 · Answer E

A 60° sector has arc length 4π. What is its radius?

A. 6 **B.** 8 **C.** 4 **D.** 24 **E.** 12

The sector is one sixth of a circle. The full circumference is 24π; dividing by 2π gives radius 12.

Question 6 · Answer A

SOLUTION **A**

Let the radius of the circle Q be q and the radius of the circle R be r. We see that the diameter of the circle P is $2q + 2r$ It follows that the radius of P is $q + r$.

We now use the formula

$$\text{circumference} = 2\pi \times \text{radius},$$

to deduce that

$$\begin{aligned}\frac{\text{circumference of } Q + \text{circumference of } R}{\text{circumference of } P} &= \frac{2\pi q + 2\pi r}{2\pi(q + r)} \\ &= \frac{2\pi(q + r)}{2\pi(q + r)} \\ &= 1\end{aligned}$$

Question 7 · Answer A

A sphere of radius 3 and a right cylinder of radius 2 have equal volumes. What is the height of the cylinder?

A. 9 **B.** 8 **C.** 12 **D.** 6 **E.** 18

The sphere volume is 36π and the cylinder volume is $4\pi h$. Equating gives $h=9$.

Question 8 · Answer B

A line with positive slope passes through (2,1) and cuts the positive x-axis at x=1. What is the square of the distance from the origin to this line?

A. 1/4 **B.** 1/2 **C.** 2 **D.** 4 **E.** 1

The line through (1,0) and (2,1) is $x-y-1=0$. The distance is $|-1|/\sqrt{2}$, whose square is 1/2.

Question 9 · Answer E

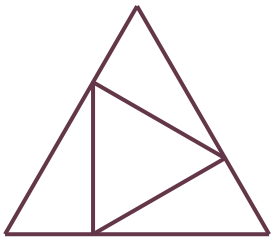
On a cube of side 6 an ant moves from one corner to the opposite corner across exactly two adjacent faces, taking a shortest route. What is the square of its route length?

A. 108 **B.** 144 **C.** 216 **D.** 324 **E.** 180

Unfold the two faces as a 12 by 6 rectangle. The straight diagonal is shortest, with squared length $144+36=180$.

Question 10 · Answer E

An equilateral triangle has side 8. A smaller equilateral triangle is formed by joining the trisection points chosen one third of the way around each side in cyclic order. What fraction of the large triangle's area is inside the smaller?



A. 2/3 **B.** 1/9 **C.** 2/9 **D.** 4/9 **E.** 1/3

Each removed corner triangle has adjacent side lengths in the proportions 1/3 and 2/3 of the original pair with the same included angle, so each occupies 2/9 of the area. Subtract 3×2/9 from 1 to get 1/3.

Question 11 · Answer A

A rectangle of perimeter 30 has integer side lengths. Among all such rectangles, what is the difference between the greatest and least possible areas?

A. 42 **B.** 49 **C.** 56 **D.** 28 **E.** 35

The side sum is 15. Closest sides 7,8 give area 56; most unequal sides 1,14 give 14. The difference is 42.

Question 12 · Answer A

Two similar solid pyramids have volumes in ratio 27:125. The larger pyramid has surface area 200. What is the surface area of the smaller?

A. 72 **B.** 90 **C.** 120 **D.** 54 **E.** 108

The length ratio is the cube root, 3:5. Areas scale as 9:25, so 200×9/25=72.

Question 13 · Answer A

A circle of radius 5 is centred at (0,0). The line y=3 cuts the circle at A and B. What is the area of triangle OAB?

A. 12 **B.** 16 **C.** 8 **D.** 24 **E.** 6

The intersection coordinates have $x=\pm4$ because $x^2+9=25$. AB has length 8 and altitude from O is 3, giving area 12.

Question 14 · Answer B

SOLUTION **B**

We note first that, by Pythagoras' Theorem applied to the right-angled triangle PQR,

$$QR^2 = PR^2 - PQ^2 = 25^2 - 20^2 = 625 - 400 = 225 = 15^2.$$

Therefore $QR = 15$.

> Note that PQR is the standard right-angled triangle with sides of lengths 3, 4 and 5 scaled up by the factor 5.
>
> If you notice that $PQ : PR = 4 : 5$, you could deduce that $QR = 15$ without the need to do the above calculation.

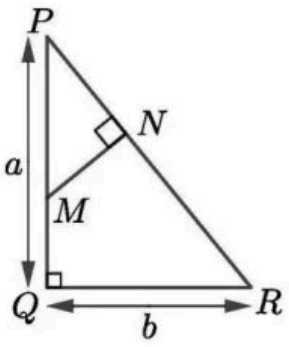
In the triangles PQR and PNM, we have $\angle PQR = \angle PNM = 90°$. These triangles share the angle at P. Because the angles in a triangle have sum 180°, it follows that $\angle PRQ = \angle NMP$. Therefore these triangles are similar.

The areas of similar triangles are proportional to the squares of the lengths of their corresponding sides. [Problem 11.2 asks you to prove this.]

Therefore since area of PNM : area of $PQR = 1 : 2$, it follows that $MN : QR = 1 : \sqrt{2}$.
Therefore $MN = \dfrac{QR}{\sqrt{2}} = \dfrac{15}{\sqrt{2}} = \dfrac{15}{2}\sqrt{2}$.

FOR INVESTIGATION

The diagram on the right shows a triangle PQR in which $\angle PQR = 90°$, $PQ = a$ and $QR = b$. M is a point on PQ and N is the point on PR such that MN is perpendicular to PR.



(a) Suppose that area of $PMN = k\times$ area of PQR, with $0 < k < 1$.

Find the ratio $PN : NR$ in terms of a, b and k.

(b) Suppose now that the point M coincides with the point Q.

Find, in terms of a and b, the ratio $PN : NR$.

Prove that the areas of similar triangles is proportional to the squares of their corresponding sides.

Question 15 · Answer D

SOLUTION **D**

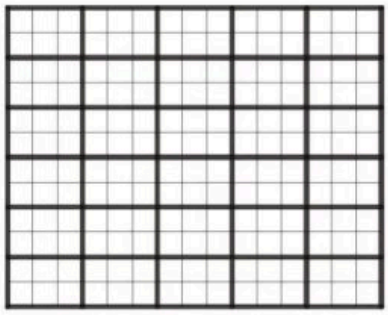
A rectangle whose sides are in the ratio 5 : 4 has dimensions $5k$ cm $\times$ $4k$ cm, for some positive number k. Since we aim to cover this rectangle with 2 cm $\times$ 3 cm rectangles without overlap, k needs to be a positive integer.

The area of the $5k$ cm $\times 4k$ cm rectangle is $(5k \times 4k)\,\text{cm}^2 = 20k^2\,\text{cm}^2$. The 2 cm $\times$ 3 cm rectangles have area $6\,\text{cm}^2$.

It follows that $20k^2$ needs to be a multiple of 6. The least positive integer k for which this is the case is 3. In this case the larger rectangle has dimensions 15 cm $\times$ 12 cm and area $(15 \times 12)\,\text{cm}^2 = 180\,\text{cm}^2$. Therefore the smallest number of 2 cm $\times$ 3 cm rectangles that are needed would be $\frac{180}{6} = 30$, provided that it is possible to cover a 15 cm $\times$ 12 cm with 30 2 cm $\times$ 3 cm rectangles.

To complete the question we need to show that this is possible.

One way in which this can be done is shown in the diagram on the right.



Therefore the smallest number of 2 cm $\times$ 3 cm rectangles that are needed is 30.

FOR INVESTIGATION

Find other ways to fit together 30 rectangles measuring 2 cm $\times$ 3 cm to make a 15 cm $\times$ 12 cm rectangle.

Question 16 · Answer E

A right triangle has hypotenuse 25 and one leg 7. A square is drawn on its other leg. What is the area of that square?

A. 625 **B.** 324 **C.** 484 **D.** 600 **E.** 576

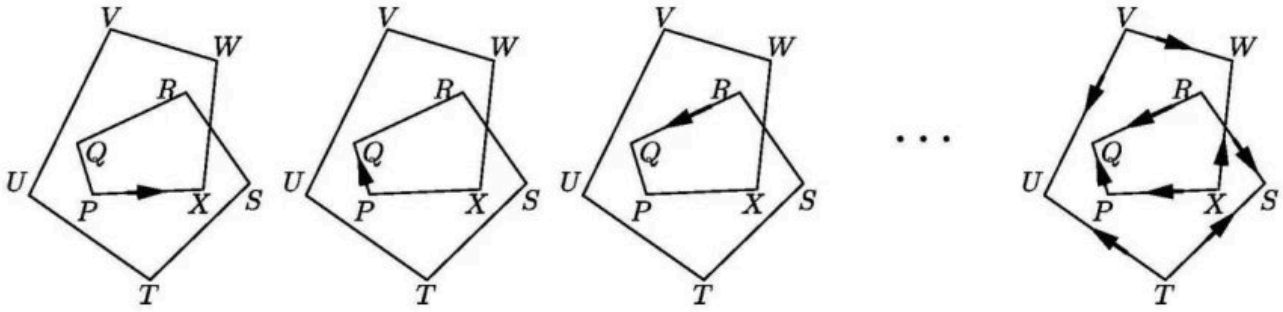
The square's area is the unknown leg squared. Pythagoras gives $625 - 49 = 576$.

Question 17 · Answer A

SOLUTION **A**

METHOD 1

We label the vertices of the figure as shown below.



We place an arrow lying along PX in the direction shown in the first figure on the left above. We rotate this arrow anticlockwise about the point P until it lies along PQ in the direction shown in the second figure above. The arrow has been turned anticlockwise through the angle $x°$.

Next we rotate the arrow anticlockwise about the point Q until it lies along RQ in the direction shown in the third figure above. As all the angles marked • are $x°$, the arrow has again been turned anticlockwise through $x°$.

We continue this process, rotating the arrow anticlockwise through $x°$ about the points R, S, T, U, V, W and X in turn. The arrow ends up lying along XP in the direction shown in the figure on the right above.

In this figure we have also shown the direction in which the arrow points on all the other edges during this process, apart from its initial position.

It will be seen that in this process the arrow has been turned through $2\frac{1}{2}$ complete revolutions. Therefore the total angle it has turned through is $2\frac{1}{2} \times 360°$, that is, through $900°$. In the process the arrow been rotated 9 times through the angle $x°$.

Therefore $9x = 900$ and hence $x = 100$.

Question 18 · Answer B

The bases of a trapezoid are 6 and 14, and its height is 9. A line parallel to the bases is drawn one third of the height up from the shorter base. What is the length of that line?

A. 10 **B.** 26/3 **C.** 8 **D.** 9 **E.** 34/3

Width changes linearly between the bases. One third of the way from 6 toward 14 adds 8/3, giving 26/3.

Question 19 · Answer D

The diagonals of a kite are perpendicular and measure 14 and 18. An enlarged copy has perimeter twice the original. What is the area of the copy?

A. 126 **B.** 252 **C.** 378 **D.** 504 **E.** 1008

The original area is half the product of the diagonals, 126. Doubling perimeter doubles every length and quadruples area, for 504.

Question 20 · Answer A

A circular pond of radius 4 has a path of constant width around it. If the path has the same area as the pond, what is the square of the outside radius?

A. 32 **B.** 64 **C.** 24 **D.** 16 **E.** 20

The total disk has area twice the pond area, or 32π. Therefore its radius squared is 32.

Question 21 · Answer D

A parallelogram has adjacent side vectors (4,1) and (1,3). What is the sum of the squares of the lengths of its two diagonals?

A. 50 **B.** 58 **C.** 40 **D.** 54 **E.** 27

The parallelogram law gives twice the sum of squared side lengths: 2[(16+1)+(1+9)]=54.

Question 22 · Answer C

SOLUTION **C**

Because Jenny and Vicky meet after walking at the same speed, they walk the same distance. Therefore

$$XL + LM = XN + NM. \qquad (1)$$

The question tells us that $XL = p$, $LM = r$ and $XN = q$, To find NM, we apply Pythagoras' Theorem to the right-angled triangle LMN. This gives

$$NM^2 = LN^2 + LM^2 = (p+q)^2 + r^2.$$

It follows that $NM = \sqrt{(p+q)^2 + r^2}$. Substituting these values in equation (1) gives

$$p + r = q + \sqrt{(p+q)^2 + r^2}.$$

Hence

$$p + r - q = \sqrt{(p+q)^2 + r^2}. \qquad (2)$$

By squaring both sides of equation (2) we obtain

$$(p + r - q)^2 = (p+q)^2 + r^2.$$

It follows that

$$p^2 + r^2 + q^2 + 2pr - 2pq - 2rq = p^2 + 2pq + q^2 + r^2. \qquad (3)$$

Equation (3) may be rearranged to give

$$4pq + 2rq = 2pr$$

from which it follows that

$$q(2p + r) = pr.$$

Therefore

$$q = \frac{pr}{2p + r}.$$

FOR INVESTIGATION

- Check that $(p + r - q)^2 = p^2 + r^2 + q^2 + 2pr - 2pq - 2rq$.
- Suppose that in this problem, $LM = LN$. In this case what is the ratio $LX : XN$?

Question 23 · Answer A

The medians of a triangle meet at G. If the region formed by G and one entire side has area 18, what is the area of the full triangle?

A. 54 **B.** 60 **C.** 45 **D.** 36 **E.** 72

The centroid is one third of the altitude above the given base, because it lies one third of the way from the base midpoint toward the opposite vertex. The smaller triangle has one third the full area. Thus the area is 54.

Question 24 · Answer D

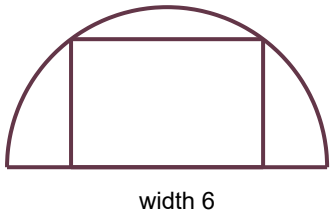
Two towers stand on level ground 12 units apart. Their heights are 5 and 14. What is the length of the straight cable joining their tops?

A. 14 **B.** 21 **C.** 13 **D.** 15 **E.** 17

The horizontal separation is 12 and the height difference is 9. Pythagoras gives √(144+81)=15.

Question 25 · Answer D

A rectangle is inscribed in a semicircle of radius 5 with its lower side on the diameter, and its two upper corners on the arc. If its width is 6, what is its area?



A. 20 **B.** 18 **C.** 12 **D.** 24 **E.** 30

The upper corners are symmetric, each horizontally 3 from the centre. The radius forms a 3–4–5 triangle, so height is 4. Area=6×4=24.